

CreditPathAI – Predicting Borrower Default Risk and Recovery Recommendations

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Abstract

CreditPathAI is a machine learning-driven platform designed to predict borrower default risk and recommend personalized recovery actions. The system leverages diverse datasets, feature engineering, and classification algorithms to provide actionable insights for collection agents. By combining predictive modeling with visualization dashboards, CreditPathAI enhances delinquency recovery efficiency and supports financial institutions in optimizing loan recovery strategies.

1. Introduction

- Importance of credit risk prediction in banking and NBFCs.
- Problem: High loan default rates and inefficiencies in manual recovery strategies.
- Objective: Build an ML platform to predict defaults and recommend recovery actions.

2. Literature

- Traditional credit scoring models (Logistic Regression, Decision Trees).
- Recent works using ensemble learning and deep learning.
- Identified gaps: lack of actionable recovery recommendations, focus only on prediction.

3. Methodology

a. Database

- Data sources: Kaggle Loan Default Dataset, Microsoft R Server Loan Credit Risk.
- Tools: Pandas, Matplotlib, Python, StreamLit.
- Features: numerical (income, loan amount, installment), categorical (employment type, credit history).

b. Preprocessing

- Handling missing values and outliers.
- Encoding categorical variables.
- Normalization of numerical features.
- Train-test split for evaluation.

c. Novelty

- Integration of prediction with personalized recovery recommendation engine.
- Actionable dashboards for loan collection agents.

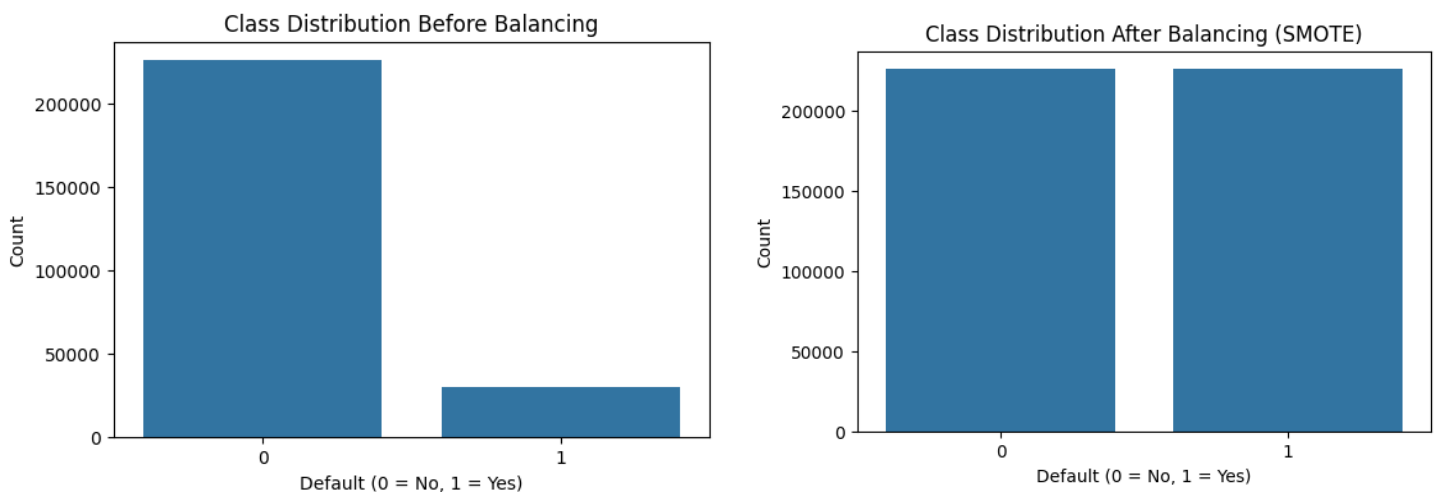
d. Proposed Method

- Classification models: Logistic Regression, Random Forest, Gradient Boosting, XGBoost, KNN.
- Evaluation using Accuracy, Precision, Recall, F1-score, ROC-AUC.

4. Results

a. EDA

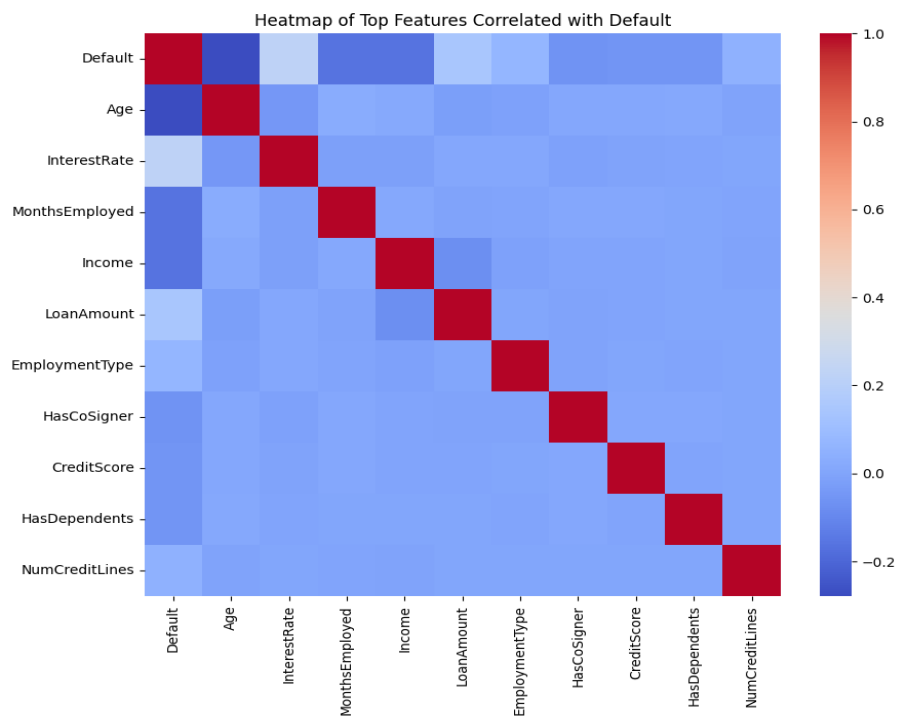
The dataset exhibited class imbalance between default and non-default borrowers. Correlation analysis highlighted income, loan amount, and installment as strong predictors.



b. Visualizations

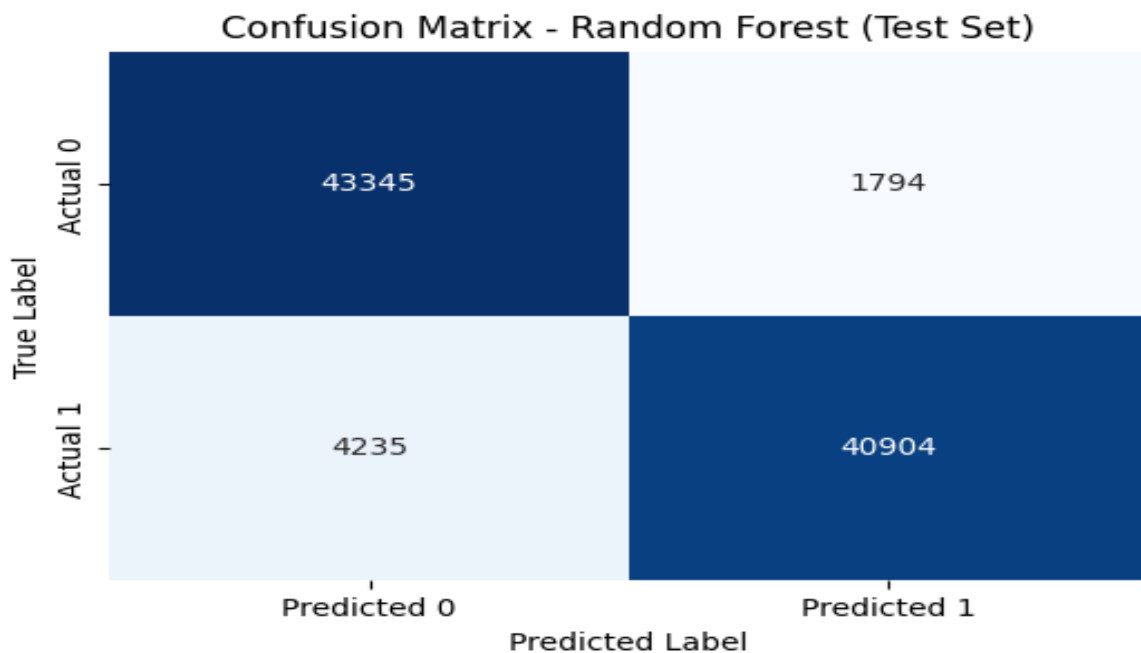
Feature distributions showed skewness in financial features. Heatmaps revealed strong correlation between credit history and default probability.

Feature importance plots emphasized credit history, income, and loan installment as key.



c. Confusion Matrices

All models produced relatively high accuracy, but trade-offs between sensitivity (recall) and specificity were evident. KNN exhibited very high sensitivity (0.99) but at the cost of lower specificity (0.62).



d. Results (Performance Metrics)

Model	Test Acc	Sensitivity	Specificity	Precision	F1-Score	AUC
Decision Tree	0.8382	0.8528	0.8236	0.8286	0.8406	0.8382
Random Forest	0.9326	0.9048	0.9603	0.9580	0.9307	0.9326
Gradient Boosting	0.9075	0.8374	0.9775	0.9739	0.9005	0.9075
XGBoost	0.9288	0.8675	0.9901	0.9887	0.9241	0.9288
KNN	0.8093	0.9900	0.6285	0.7272	0.8385	0.8093

5. Discussion

- Random Forest achieved the highest overall test accuracy (0.9326) with a strong balance between sensitivity (0.90) and specificity (0.96).
- XGBoost closely followed (accuracy 0.9288) but stood out with the highest specificity (0.99) and precision (0.99).
- Gradient Boosting delivered good accuracy (0.9075) and the second-best specificity (0.98), making it highly reliable in avoiding false approvals.
- Decision Tree performed moderately (accuracy 0.8382). While interpretable, its lower metrics compared to ensemble models indicate overfitting and limited predictive power.
- KNN showed very high sensitivity (0.99), meaning it almost never missed a default case. However, its low specificity (0.62) and moderate precision (0.72) resulted in too many false positives.
- Overall, ensemble models (Random Forest, Gradient Boosting, XGBoost) outperformed single learners (Decision Tree, KNN). Among them, Random Forest provided the most balanced and reliable results, while XGBoost was the most conservative in flagging defaults.

6. Conclusion

CreditPathAI successfully predicts borrower default risk and integrates a recovery recommendation system. With strong performance on benchmark datasets, the system demonstrates practical applicability for banks and NBFCs.

7. Future Works

- Integration with real-time financial data streams.
- Deployment as a web-based dashboard with API support using Flask, React.
- Incorporating explainable AI (SHAP, LIME) for regulatory compliance.
- Expanding datasets across geographies for generalization.